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EE208 Control Engineering Lab Report
Experiment 4

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1. Aim

The aim of this experiment is to evaluate the performance of a Mitsubishi 230 V AC servomotor, controlled by a Mitsubishi MR-JE-40AS servo controller, operating in both speed and position control modes.

2. Objective

The objectives of the experiment are:

- To study the fundamental operation of a closed-loop servomotor actuator that utilizes a photoelectric sensor and position feedback to control its motion.
- To evaluate the performance of the servomotor under a bang-bang control algorithm by setting different reference speeds (within a 2000 rpm limit) and analyzing the resulting percentage error trends.
- To assess the accuracy of the servomotor's position control by applying various angular positions.
- To analyze the operational constraints of the servomotor and discuss the necessity of strictly adhering to the prescribed speed and position limits.

3. Applications of Control in Servomotors

Servomotors, governed by closed-loop control strategies such as the one studied in this experiment, are used extensively across industry wherever precise, repeatable motion is required. A few representative applications are summarized below; a more detailed treatment is given in Section 8.

- **Automated Flow Control Valves:** Servomotors are used to precisely throttle and regulate fluid flow by rotating and holding valves at specific fractional angles.
- **CNC Machinery and 3D Printers:** Servomotors drive the X, Y, and Z axes through ball screws in milling machines, laser cutters, lathes, and 3D printers, enabling precise tool positioning.
- **Robotic Joints and Manipulators:** Servomotors actuate multi-axis movements in industrial robotic arms used for applications such as welding, pick-and-place operations, and automated assembly.
- **Aerospace and Defense:** High-reliability servo actuators are used for applications such as fly-by-wire flight control surfaces, radar dish positioning, and thrust-vectoring mechanisms.

4. Bang-Bang Control of a Servo Motor

A bang-bang controller, also known as an ON–OFF controller, is one of the simplest control strategies used for controlling the speed or position of a servo motor. Unlike a PID controller, which continuously changes its output according to the magnitude of the error, a bang-bang controller generally provides only a few discrete control commands, such as maximum positive, zero, or maximum negative control.

For a servo motor, bang-bang control is particularly useful when the objective is to move the shaft quickly from one position to another and stop it at the desired position.

$$u(t) = \begin{cases} +U_{max}, & e(t) > 0 \\ -U_{max}, & e(t) < 0 \end{cases}$$

where U_{max} is the maximum available control command.

In practice, a deadband or hysteresis is normally introduced to prevent rapid switching near the reference position:

$$u(t) = \begin{cases} +U_{max}, & e(t) > e_d \\ 0, & |e(t)| \leq e_d \\ -U_{max}, & e(t) < -e_d \end{cases}$$

where e_d is the deadband.

4.3 Transfer Characteristics of the Bang-Bang Controller

The transfer characteristics of a bang-bang controller illustrate the relationship between the error signal and the controller output. In the ideal case, the controller switches instantaneously between the positive and negative maximum control levels when the error changes its sign. In practical systems, a dead zone or hysteresis is introduced around the zero-error region to prevent rapid switching and chattering.

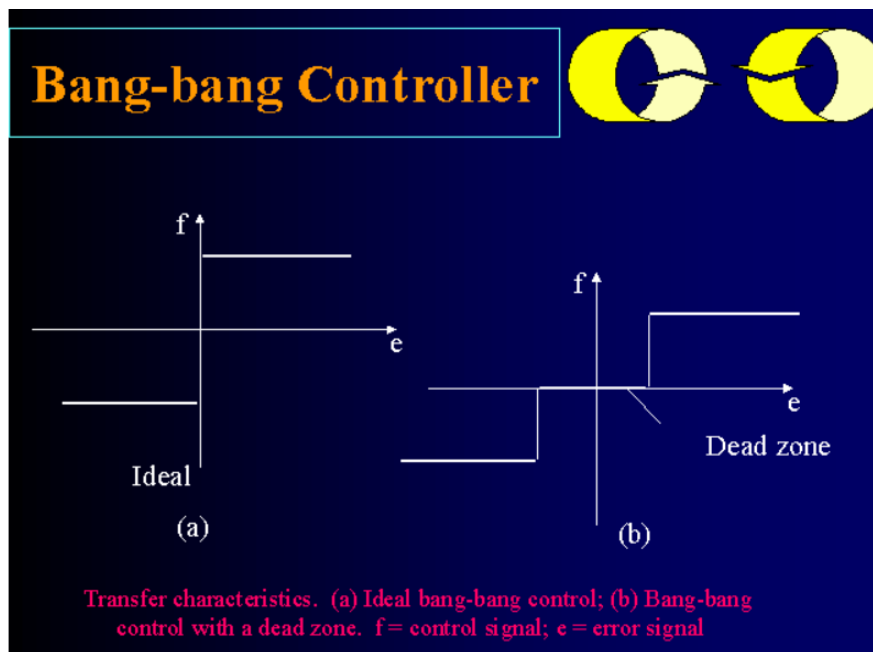


Figure 2: Transfer characteristics of an ideal bang-bang controller and a bang-bang controller with dead zone.

For the servo motor used in this experiment, the error signal is defined as

$$e\theta = \theta_{ref} - \theta_{actual}$$

The ideal bang-bang controller can be represented as

$$u(e_\theta) = \begin{cases} +U_{max}, & e_\theta > 0 \\ -U_{max}, & e_\theta < 0 \end{cases}$$

However, to avoid excessive switching near the desired position, a dead zone is introduced. The practical control law is therefore

$$u(e_\theta) = \begin{cases} +U_{max}, & e_\theta > \Delta\theta \\ 0, & |e_\theta| \leq \Delta\theta \\ -U_{max}, & e_\theta < -\Delta\theta \end{cases}$$

Here, $\Delta\theta$ represents the position deadband. When the position error lies within this region, the controller output becomes zero, allowing the servomotor to settle near the desired position.

4.4 Why is it Called Bang-Bang Control?

The name bang-bang comes from the controller switching abruptly between its extreme control values. Instead of gradually increasing or decreasing the control signal, the controller applies maximum control in the required direction and changes its output when the reference condition is reached.

The controller can therefore operate as:

$$+U_{max} \rightarrow 0 \rightarrow -U_{max}$$

depending on the position error and the direction of motion.

4.5 Bang-Bang Control for Position Control

Consider a servo motor initially located at

$$\theta = 0^\circ$$

and suppose the desired position is

$$\theta_{ref} = 90^\circ$$

Initially,

$$e = 90^\circ$$

Since the error is positive, the controller applies maximum positive control. The motor therefore accelerates toward the desired position.

As the shaft approaches 90° , the position error decreases. When the motor enters the specified deadband, the controller removes or changes the control command, allowing the motor to stop near the desired position.

4.6 Typical Operating Sequence

The motion of the servo motor under bang-bang control can be divided into three stages.

Stage 1: Acceleration

Initially,

$$e \gg 0$$

Therefore,

$$u = +U_{\max}$$

The motor accelerates toward the desired position.

Stage 2: Approaching the Target

As the motor approaches the reference position,

$$e \rightarrow 0$$

The controller changes its output according to the defined switching condition.

Stage 3: Stopping

When

$$|e| \leq e_d$$

the controller gives

$$u = 0$$

Ideally, the motor then stops at or very close to the desired reference position.

4.7 Closed-Loop Block Diagram of the Servo Motor

The basic closed-loop control structure of the servo motor used in the experiment is shown in Fig. 3.

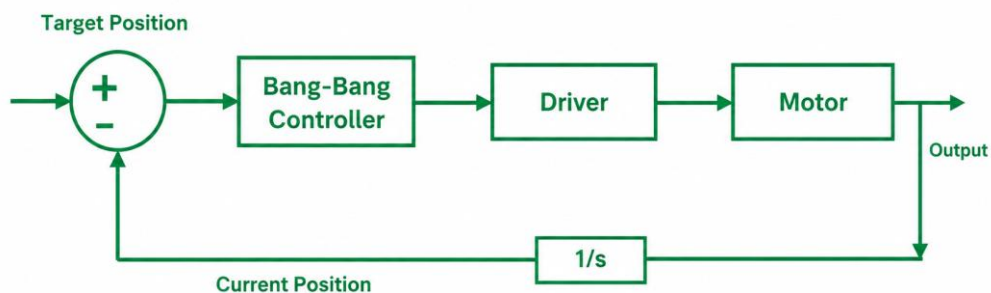


Figure 3: Closed-loop block diagram of the servo motor control system.

The servo motor operates as a closed-loop system in which the desired position is compared with the current motor position. The resulting position error is supplied to the controller, which generates the appropriate control signal. This signal is then applied to the motor through the driver.

The motor output is measured using an encoder. The encoder provides feedback corresponding to the current position of the motor shaft. This feedback is compared with the target position, thereby forming a closed-loop control system.

The position error is given by

$$e(t) = \theta_{ref}(t) - \theta(t)$$

where θ_{ref} is the target position and θ is the actual motor position obtained from the encoder.

For bang-bang control, this error is used to determine the control action applied to the motor. The controller switches its output according to the magnitude and direction of the position error.

4.8 Bang-Bang Control with Speed Feedback

A more practical servo control structure uses the bang-bang controller to generate a speed reference rather than directly controlling the motor voltage.

The position error is first calculated as

$$e_{\theta} = \theta_{ref} - \theta_{actual}$$

The bang-bang controller then generates a speed reference according to

$$\omega_{ref} = \begin{cases} +\omega_{max}, & e_{\theta} > e_d \\ 0, & |e_{\theta}| \leq e_d \\ -\omega_{max}, & e_{\theta} < -e_d \end{cases}$$

The speed reference is then compared with the actual motor speed and supplied to the speed controller.

The actual motor speed is fed back to the speed controller, while the motor speed can also be integrated to obtain position:

$$\theta(t) = \int \omega(t) dt$$

4.9 Bang-Bang Controller with Hysteresis

A major problem with a pure bang-bang controller is chattering. When the motor approaches the reference position, small variations in the measured position can cause the controller to repeatedly switch between positive and negative control commands. For example,

$$+U_{max} \rightarrow -U_{max} \rightarrow +U_{max} \rightarrow -U_{max}$$

This rapid switching can cause vibration, unnecessary switching losses, and mechanical stress.

To reduce this problem, a deadband or hysteresis is introduced. For example, if

$$e_d = 1^\circ$$

then

$$e > +1^\circ \Rightarrow +U_{max}$$

$$-1^\circ \leq e \leq +1^\circ \Rightarrow 0$$

$$e < -1^\circ \Rightarrow -U_{\max}$$

The deadband prevents unnecessary switching when the motor is sufficiently close to the desired position.

4.10 Mathematical Model of the Servo Motor

For a simplified servo motor, the mechanical equation can be written as

$$J \frac{d\omega}{dt} + B\omega = T_m - T_L$$

where:

- J is the moment of inertia.
- B is the viscous friction coefficient.
- ω is the angular velocity.
- T_m is the motor torque.
- T_L is the load torque.

The relationship between angular velocity and position is

$$d\theta / dt = \omega$$

Therefore,

$$\theta(t) = \int \omega(t) dt$$

The bang-bang controller determines the direction and magnitude of the commanded control action, while the motor dynamics determine the actual motion of the shaft.

4.11 High-Speed Overtravel and Overshoot

The operating speed of the servo motor has a significant influence on positioning accuracy. The kinetic energy of the rotating rotor is

$$E_k = \frac{1}{2} J \omega^2$$

Since kinetic energy is proportional to the square of angular velocity, a motor operating at a higher speed possesses significantly more mechanical energy.

When the controller removes the driving command near the target position, the motor may not stop immediately because of its mechanical inertia. The shaft can continue rotating beyond the reference position. Therefore,

$$\theta_{actual} > \theta_{ref}$$

which results in overshoot. For example, if

$$\theta_{ref} = 90^\circ \quad \text{and} \quad \theta_{actual} = 93^\circ$$

then

$$e\theta = 90 - 93 = -3^\circ$$

Thus, the motor has travelled beyond the desired position.

4.12 Sensor Sampling Rate Limitations

In a digital control system, the feedback signal is measured at discrete sampling intervals. If the sampling period is T_s , the approximate angular displacement between two consecutive measurements is

$$\Delta\theta \approx \omega T_s$$

At higher speed,

$$\omega \uparrow \Rightarrow \Delta\theta \uparrow$$

Therefore, the motor can travel a larger angular distance between two consecutive sensor measurements. This effect can increase the position error, particularly for small angular targets.

4.13 Low-Speed Undershoot

At low speed, the motor possesses comparatively less kinetic energy and momentum. When the motor enters the deadband, the driving command may be removed.

However, static friction or stiction can prevent the rotor from moving further toward the reference position. As a result,

$$\theta_{actual} < \theta_{ref}$$

which produces an undershoot. For example, if

$$\theta_{ref} = 90^\circ \quad \text{and} \quad \theta_{actual} = 88.5^\circ$$

then

$$e\theta = 90 - 88.5 = 1.5^\circ$$

Therefore, the motor has stopped slightly before reaching the desired position.

4.14 Effect of Angular Range

The commanded angular displacement also affects the observed percentage error. For a small angular command, even a small absolute error can represent a large percentage of the reference value.

For example, if the absolute position error is approximately 2° , then for a 20° command:

$$\%Error = (2 / 20) \times 100 = 10\%$$

For a 90° command:

$$\%Error = (2 / 90) \times 100 \approx 2.22\%$$

Thus, even when the absolute stopping error is similar, the percentage error can be significantly larger for smaller commanded angular positions.

4.15 Effect of Motor Speed

The motor speed has a major influence on the positioning performance.

High-Speed Operation

Advantages:

- Faster movement.
- Shorter positioning time.
- Higher productivity.

Disadvantages:

- Higher kinetic energy.
- Greater overshoot.
- Larger stopping distance.
- Greater sensitivity to sampling delays.
- Potentially larger position error.

Low-Speed Operation

Advantages:

- Easier stopping.
- Lower overshoot.
- Better controllability near the target.

Disadvantages:

- Longer positioning time.
- Greater influence of static friction.
- Possibility of stopping before reaching the exact reference position.

4.16 Relay Control and Deadband (ϵ) Dynamics in Servomotor Actuators

A servomotor governed by bang-bang control can be modeled as a relay controller in which the control input is restricted to discrete saturation states rather than being continuously proportional to the position error. The controller produces a maximum positive control input, a maximum negative control input, or zero control input depending on the magnitude and direction of the error.

The position error is defined as

$$e_\theta = \theta_{ref} - \theta_{actual}$$

The relay control law with a deadband ϵ can be expressed as

$$u(e_\theta) = \begin{cases} V^+, & e_\theta > \epsilon \\ 0, & |e_\theta| \leq \epsilon \\ V^-, & e_\theta < -\epsilon \end{cases}$$

where V^+ represents the maximum forward control voltage, V^- represents the maximum reverse control voltage, and ε represents the allowable position-error deadband.

$$u(t) = \begin{cases} h & \text{for } e(t) \geq -\varepsilon \\ -h & \text{for } e(t) < -\varepsilon \end{cases}$$

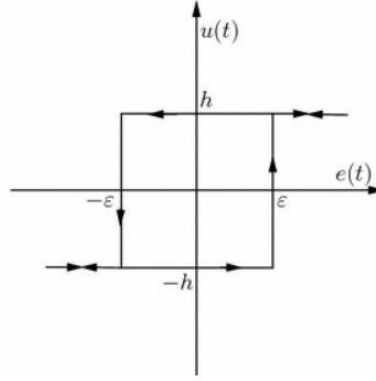


Figure 4: Relay-type bang-bang control characteristic with a deadband ε .

The deadband is introduced to prevent excessive switching, commonly known as chattering, near the reference position. Without a deadband, small measurement noise or very small changes in position error could cause the controller to repeatedly switch between V^+ and V^- . Such rapid switching can increase electrical stress, mechanical wear, and unnecessary power losses.

Therefore, the controller is designed to produce zero control action when $|e| \leq \varepsilon$. The relationship between V^+ , V^- , and ε plays an important role in determining the nonlinear dynamic response of the servomotor. The effects of this relay behaviour can be observed directly in the experimental speed and position results.

Small-Angle Positioning Error

For small position commands such as 5° and 20° , the initial position error is large enough to activate the V^+ control state. Since the commanded angular distance is very small, the motor reaches the ε region quickly. However, the rotor has already acquired significant kinetic energy and cannot stop instantaneously when the control input changes to zero.

The rotational kinetic energy of the rotor is

$$E_k = \frac{1}{2} J \omega^2$$

where J is the moment of inertia and ω is the angular velocity. The stored kinetic energy causes the rotor to continue moving after entering the deadband. Consequently, the motor can travel beyond the desired position, producing a large positioning error.

This effect is particularly important for small angular commands because the stopping distance represents a large fraction of the commanded displacement. In the experiment, 5° and 20° commands at 1800 rpm resulted in approximately 80% and 95% positioning error, respectively.

Low-Speed Stiction and Deadband

At extremely low reference speeds, such as 11 rpm and 25 rpm, the available motor torque and momentum are very small. When the position or speed error approaches the deadband, the relay controller may provide short control actions. These short control pulses may not generate sufficient torque to overcome static friction, also known as stiction.

The motor can therefore remain stationary even though a small error exists. This produces a large steady-state speed error. In the experimental results, the measured speed was zero for reference speeds of 11 rpm and 25 rpm, resulting in

$$\%Error = |N_{ref} - N_{meas}| / N_{ref} \times 100 = 100\%$$

This demonstrates the practical limitation of relay-based control at extremely low operating speeds.

Large-Signal Positioning and Stable Operation

For large angular commands, such as 200°, 250°, and 330°, the motor operates for a longer duration before reaching the target region. During this interval, the controller can maintain the appropriate saturation state for a longer period.

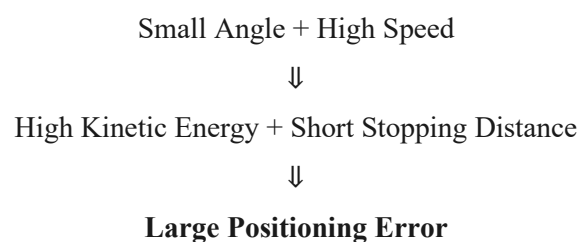
The deadband ϵ becomes relatively small compared with the total angular displacement. Consequently, the effects associated with the deadband represent a smaller fraction of the overall motion.

For example, at 1800 rpm, several large-angle forward commands such as 90°, 120°, 200°, 250°, and 330° were achieved with approximately zero forward-position error in the measured data.

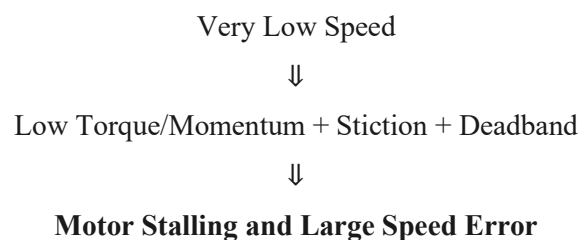
This demonstrates that the relay controller can provide good positioning performance for suitable combinations of speed and angular displacement, even though its discrete control action can produce significant errors for small-angle or extremely low-speed operation.

Relation Between Deadband and Experimental Behaviour

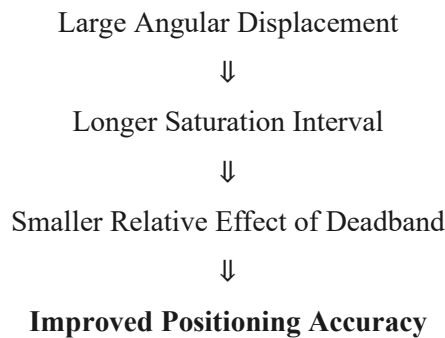
The observed experimental behaviour can therefore be related to the relay control characteristics as follows:



Similarly,



whereas for large angular commands,



Thus, the deadband ε is an important design parameter in the bang-bang controller. A very small deadband can cause excessive switching, while an excessively large deadband can reduce positioning accuracy. Therefore, an appropriate deadband must be selected to obtain a balance between switching stability and positioning accuracy.

4.17 Percentage Position Error

The percentage position error can be calculated using

$$\% \text{Error} = \frac{|\theta_{ref} - \theta_{actual}|}{|\theta_{ref}|} \times 100$$

For example, if

$$\theta_{ref} = 60^\circ \quad \text{and} \quad \theta_{actual} = 62^\circ$$

then

$$\% \text{Error} = |60 - 62| / 60 \times 100 = 3.33\%$$

This calculation can be used to compare the positioning accuracy at different speeds and commanded angular positions.

4.18 Comparison of PID and Bang-Bang Control Responses

The transient responses obtained under PID and bang-bang control demonstrate the difference in their control actions. The PID-controlled response exhibits a relatively smooth oscillatory response that gradually settles toward the steady-state value. In contrast, the bang-bang controlled response reaches the operating region rapidly with a comparatively shorter transient period and smaller sustained oscillations.

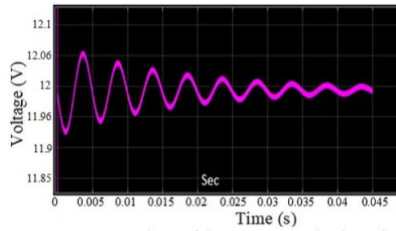


Figure 5: Output voltage of the converter under the action of a PID.

Figure 5: Output voltage of the converter under PID control.

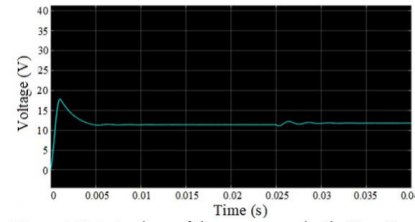


Figure 6: Output voltage of the converter under the Bang-Bang control action.

Figure 6: Output voltage of the converter under bang-bang control.

From Fig. 5, the PID-controlled output shows an oscillatory transient response with the output gradually settling around its steady-state value. The oscillations decrease with time, indicating a damped response.

In comparison, Fig. 6 shows the response under bang-bang control. The output reaches the steady-state region rapidly, although a small disturbance or oscillation can still be observed during the transient period.

The comparison demonstrates that the bang-bang controller can provide a fast response with a simple control structure, whereas PID control provides a smoother and more continuously regulated response. The choice between the two controllers depends on the required settling time, accuracy, complexity, and acceptable overshoot.

Feature	Bang-Bang	PID
Control output	Discrete	Continuous
Complexity	Low	Higher
Response	Fast	Tunable
Overshoot	Can be significant	Can be reduced
Precision	Moderate	High
Chattering	Possible	Generally lower
Tuning	Minimal	Requires Kp, Ki, Kd

Table 1: Comparison of PID and Bang-Bang Control.

4.19 Bang-Bang Control in the Present Experiment

For the Mitsubishi AC servomotor experiment, the overall control concept can be represented as

Reference Speed/Position → Bang-Bang Controller → Servo Controller → AC Servo Motor → Photoelectric Sensor (Feedback)

The controller compares the reference quantity with the feedback signal and generates the appropriate control action.

The main experimental relationship being studied is

Reference → Bang-Bang Control → Motor Motion → Feedback → Position/Speed Error

During the experiment, the reference speed can be varied within the prescribed limit of 2000 rpm, and different angular positions can be commanded. The resulting actual position and percentage error can then be recorded and compared.

4.20 Photoelectric Sensor in Servo Motor

Working Principle

A photoelectric sensor used in a servo motor commonly employs a phototransistor to convert incident light into an electrical signal. A phototransistor is a light-sensitive semiconductor device in which the incident photons generate electron–hole pairs, causing the collector current to vary with the intensity of the incident light.

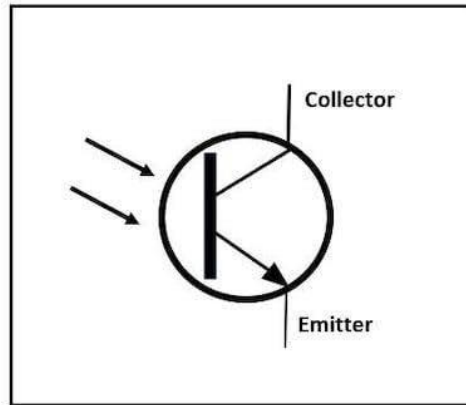


Figure 7: Symbolic representation of a phototransistor.

When light falls on the semiconductor material, photons with sufficient energy excite electrons from the valence band to the conduction band, generating additional charge carriers. The resulting change in carrier concentration produces a change in the collector current, which can be detected by the servo controller as an electrical feedback signal.

In the semiconductor junction, **the Fermi-level difference between the different semiconductor regions establishes the built-in potential and affects the movement and distribution of charge carriers.** Under illumination, the generation of additional electron–hole pairs modifies the carrier distribution and consequently changes the electrical current through the device. This change in current is used to detect the presence or interruption of light.

In the servo motor system, a rotating encoder disc periodically blocks and allows the light reaching the phototransistor. This produces a sequence of electrical pulses corresponding to the rotation of the motor shaft. These pulses are processed by the servo controller to determine the actual position and speed of the motor.

Sensing Errors and Limitations

The accuracy of the photoelectric sensor can be affected by several factors such as limited resolution, sampling delay, missed pulses, electrical noise, misalignment, and variations in the intensity of the light source. At high speeds, the shaft can rotate through a larger angle between two consecutive measurements, causing the controller to detect the target position slightly late. This can result in over-travel and positioning errors.

The sensor also has limitations in detecting very small angular movements. Therefore, for small commanded angles, even a small sensing error can produce a relatively large percentage error. At low speeds, friction, deadband, and sensor resolution can further affect the accuracy of position detection.

The position error is given by

$$e\theta = \theta_{ref} - \theta_{actual}$$

The percentage positioning error is calculated as

$$\% \text{Error} = \frac{|\theta_{ref} - \theta_{actual}|}{|\theta_{ref}|} \times 100$$

The main disadvantages affecting sensing accuracy are:

- **Limited resolution:** Small angular movements may not be detected accurately.
- **Sampling delay:** Delay between sensing and controller response can cause over-travel at high speeds.
- **Missed pulses:** At high rotational speeds, pulses may be missed, leading to incorrect position or speed estimation.
- **Electrical noise:** Noise can introduce false or distorted feedback signals.
- **Misalignment:** Improper alignment of the light source, encoder disc, and detector can reduce sensing accuracy.
- **Light interference:** External light sources can interfere with optical detection.
- **Small-angle error:** A small absolute sensing error can produce a large percentage error when the commanded angular displacement is small.

Thus, the accuracy of the photoelectric feedback system directly affects the overall positioning accuracy and performance of the servo motor.

4.21 Overall Summary

The bang-bang controller provides a simple and effective method for controlling a servo motor by switching between maximum control actions according to the position error. Its performance is strongly affected by motor speed, inertia, friction, sensor sampling, deadband, and commanded angular displacement.

In the present experiment, higher operating speeds tend to produce greater overshoot because of the increased kinetic energy of the rotor. At lower speeds, static friction and deadband can become more significant and may cause the motor to stop slightly before the desired position. Furthermore, smaller angular commands can exhibit larger percentage errors because the same absolute positioning error represents a larger fraction of a smaller reference angle.

Thus, bang-bang control provides a simple and fast control strategy, but highly precise servo positioning generally requires additional techniques such as hysteresis, braking control, speed feedback, or a PID-based inner control loop.

5. Experimental Results and Analysis

The experimental investigation was performed to study the speed and position control performance of the Mitsubishi AC servomotor under bang-bang control. The analysis was divided into position control at constant speed, reverse speed control, and forward speed control. The errors were calculated by comparing the measured values with their corresponding reference values.

The signed error is defined as

$$e = \text{Measured Value} - \text{Reference Value}$$

A positive value of e indicates that the measured value is greater than the reference value, while a negative value indicates that the measured value is less than the reference value.

The key thing we observed is that at high speed more error is observed when angle is lower ones and when we operate our servo over lower speed ranges the more error is at lower angle of operations in most of the cases.

5.1 Position Control at Constant Speed

The position-control experiment was performed at different constant speed settings. Different angular positions were commanded, and the corresponding forward and backward observed angles were recorded. For the error analysis, only the first observed value, i.e., the forward observed angle, is considered.

The angular position error is calculated as

$$e_{\theta} = \theta_{FW,obs} - \theta_{set}$$

and the corresponding percentage error is

$$\% \text{Position Error} = \frac{|e_{\theta}|}{\theta_{set}} \times 100$$

A positive angular error indicates over-travel, whereas a negative angular error indicates that the motor stopped before reaching the commanded position.

Speed (rpm)	Set Angle (°)	FW Obs. (°)	BW Obs. (°)	FW Error (°)	Error (%)	FW Steps
1800	5	1	4	-4	80.00	5
1800	35	35	0	0	0.00	1
1800	20	1	19	-19	95.00	19
1800	120	120	0	0	0.00	1
1800	90	90	0	0	0.00	1
1800	200	200	0	0	0.00	1
1800	250	250	0	0	0.00	1
1800	330	330	152	0	0.00	1
800	50	50	0	0	0.00	1
800	100	100	0	0	0.00	1
800	150	150	0	0	0.00	1
800	200	200	0	0	0.00	1
800	300	300	116	0	0.00	1
800	330	330	0	0	0.00	1
800	30	27	3	-3	10.00	4
800	15	1	14	-14	93.33	13

Speed (rpm)	Set Angle (°)	FW Obs. (°)	BW Obs. (°)	FW Error (°)	Error (%)	FW Steps
100	30	30	0	0	0.00	1
100	130	115	15	-15	11.54	2
100	100	100	0	0	0.00	1
100	83	83	0	0	0.00	1
100	65	55	0	-10	15.38	2
100	200	116	85	-84	42.00	2
100	280	176	42	-104	37.14	2
100	330	116	214	-214	64.85	3
50	30	30	0	0	0.00	1
50	60	59	32	-1	1.67	2
50	100	59	71	-41	41.00	2
50	150	59	91	-91	60.67	3
50	200	59	141	-141	70.50	3
50	250	59	221	-191	76.40	5
50	40	40	0	0	0.00	1
50	5	5	0	0	0.00	1

Table 2: Position control results at constant speed.

Position Control Analysis

The position-control results show that the error depends strongly on both the reference speed and the commanded angular displacement. At 1800 rpm, very small angular commands produced significant errors. For example, for a 5° command, the observed forward angle was only 1° , giving

$$e\theta = 1 - 5 = -4^\circ \quad \text{and} \quad \% \text{Error} = 4 / 5 \times 100 = 80\%$$

For a 20° command, the observed angle was 1° , resulting in

$$e\theta = 1 - 20 = -19^\circ \quad \text{and} \quad \% \text{Error} = 95\%$$

In contrast, several larger angular commands at 800 rpm and 1800 rpm were achieved with negligible forward-position error. Therefore, the data do not show a simple relationship in which higher speed always produces greater error. Instead, small-angle positioning is particularly sensitive to the stopping dynamics, sensor response, and control deadband.

At lower speeds, significant errors were also observed. For example, at 50 rpm, a 250° command resulted in a forward observed angle of 59° , giving a negative error of

$$e\theta = 59 - 250 = -191^\circ$$

corresponding to a percentage error of

$$\% \text{Error} = 191 / 250 \times 100 = 76.4\%$$

This indicates the increasing influence of friction, stiction, and deadband at low operating speeds.

5.2 Reverse Speed Control Analysis

The reverse-speed experiment was performed by applying different reference speed commands and measuring the corresponding reverse motor speed. When two measurements were recorded for a particular reference speed, both values are included in the table.

The signed speed error is calculated as

$$eN = N_{\text{measured}} - N_{\text{reference}}$$

The percentage speed error is

$$\% \text{Speed Error} = \frac{|N_{\text{measured}} - N_{\text{reference}}|}{N_{\text{reference}}} \times 100$$

Sr. No.	Ref. Speed (rpm)	Measured Speed (rpm)	Signed Error (rpm)	Error (%)
1	950	950/947	0/-3	0/0.32
2	1000	1006/997	+6/-3	0.60/0.30
3	1100	1103/993	+3/-107	0.27/9.73
4	1200	1206/1196	+6/-4	0.50/0.33
5	1300	1307/1292	+7/-8	0.54/0.62
6	1400	1403/1386	+3/-14	0.21/1.00
7	1500	1516/1496	+16/-4	1.07/0.27
8	1600	1601	+1	0.06
9	1700	1693	-7	0.41
10	1800	1805	+5	0.28
11	1900	1896	-4	0.21
12	2000	1996	-4	0.20

Table 3: Reverse speed control results and errors.

Reverse Speed Analysis

The reverse-speed measurements show that the servo motor generally tracks the reference speed accurately at medium and high operating speeds. Most measured values differ from the reference by only a few rpm.

For example, at 2000 rpm, the measured speed was 1996 rpm, giving

$$eN = 1996 - 2000 = -4 \text{ rpm} \quad \text{and} \quad \% \text{Error} = 0.20\%$$

At 1800 rpm, the measured speed was 1805 rpm:

$$eN = 1805 - 1800 = +5 \text{ rpm}, \quad \text{giving } \% \text{Error} \approx 0.28\%$$

The presence of both positive and negative errors indicates that the measured speed fluctuates slightly around the reference value rather than showing a consistent tendency to overestimate or underestimate the speed.

5.3 Forward Speed Control Analysis

The forward-speed experiment was conducted over a wider range, including very low speeds as well as speeds approaching the 2000 rpm operating limit.

Sr. No.	Ref. Speed (rpm)	Measured Speed (rpm)	Signed Error (rpm)	Error (%)
1	11	0	-11	100.00
2	100	100	0	0.00
3	50	50	0	0.00
4	25	0	-25	100.00
5	35	35	0	0.00
6	1000	1004/995	+4/-5	0.40/0.50
7	330	331	+1	0.30
8	350	350/349	0/-1	0.00/0.29
9	400	400/401	0/+1	0.00/0.25
10	450	450/448	0/-2	0.00/0.44
11	500	500	0	0.00
12	550	551/448	+1/-102	0.18/18.55
13	600	601/597	+1/-3	0.17/0.50
14	650	652/649	+2/-1	0.31/0.15
15	700	701/699	+1/-1	0.14/0.14
16	800	800	0	0.00
17	950	955/947	+5/-3	0.53/0.32
18	1000	1006/997	+6/-3	0.60/0.30
19	1100	1103/1093	+3/-7	0.27/0.64
20	1200	1206/1196	+6/-4	0.50/0.33
21	1300	1307/1292	+7/-8	0.54/0.62
22	1400	1404/1386	+4/-14	0.29/1.00
23	1500	1496/1516	-4/+16	0.27/1.07
24	1600	1601	+1	0.06
25	1800	1805/1777	+5/-23	0.28/1.28
26	2000	2003/1996	+3/-4	0.15/0.20

Table 4: Forward speed control results and errors.

Forward Speed Analysis

The forward-speed results demonstrate two distinct operating regions. At very low reference speeds, the motor may fail to achieve the commanded speed. For example, at 11 rpm and 25 rpm, the measured speed was zero. The corresponding errors are

$$eN = 0 - 11 = -11 \text{ rpm} \quad \text{and} \quad eN = 0 - 25 = -25 \text{ rpm}$$

corresponding to 100% error in both cases. This behaviour can be attributed to the minimum effective torque, static friction, deadband, and limitations of the sensing and control system at very low speeds.

For medium and high speeds, the tracking accuracy improves considerably. For example, at 800 rpm the measured speed was exactly 800 rpm. At 1600 rpm the measured speed was 1601 rpm, giving an error of only

$$eN = 1601 - 1600 = +1 \text{ rpm}$$

At 2000 rpm, the measured values were 2003 rpm and 1996 rpm, corresponding to $eN = +3$ rpm and $eN = -4$ rpm, respectively. Thus, the servo motor maintains good speed tracking close to its maximum specified operating speed.

5.4 Comparative Analysis of Speed and Position Control

The experimental results reveal an important distinction between speed accuracy and position accuracy. The speed-control measurements are generally very close to their reference values at medium and high speeds. However, the position-control results can exhibit substantial errors for certain angular commands.

This indicates that accurate speed tracking does not necessarily guarantee accurate positioning. Positioning accuracy additionally depends on the stopping dynamics of the motor, mechanical inertia, friction, sensor resolution, controller deadband, and response delay.

The high-speed position-control results particularly demonstrate the effect of stopping dynamics. For example, at 1800 rpm, a 5° reference resulted in a -4° forward error, corresponding to an 80% percentage error. However, a 90° reference at the same speed resulted in zero forward error.

At low speed, friction and stiction become more important. At 50 rpm, the 250° command resulted in a -191° forward error, corresponding to a 76.4% error.

Therefore, the experimental results can be summarized as

$$\text{Speed Accuracy} \neq \text{Position Accuracy}$$

and

$$\text{Position Accuracy} = f(\text{Speed, Angle, Inertia, Friction, Deadband, Feedback})$$

5.5 Overall Experimental Observation

The key observation from the experiment is that the positioning error is strongly dependent on both speed and angular displacement. At high speeds, small-angle commands are particularly sensitive to stopping delay, rotor inertia, and discrete sensor feedback. At low speeds, the influence of friction, stiction, and deadband becomes more significant.

The speed-control experiment, however, shows that the servo motor tracks medium and high reference speeds with relatively small error. Therefore, the large positioning errors observed in some cases are mainly associated with the positioning and stopping behaviour rather than poor steady-state speed regulation.

The experimental results demonstrate that:

- High-speed operation can increase the effect of rotor inertia and stopping delay, particularly for small angular commands.

- Small angular commands can produce large percentage errors because even a small absolute error represents a large fraction of the reference angle.
- Low-speed operation increases the relative influence of friction, stiction, and controller deadband.
- The speed measurements remain close to their reference values over most of the medium and high speed range.
- Positive and negative speed errors occur, indicating small fluctuations around the reference speed rather than a constant bias.
- Accurate speed control does not necessarily imply accurate position control because positioning also depends on the stopping and feedback dynamics.

Thus, the optimum operating condition for accurate positioning requires an appropriate combination of reference speed, angular displacement, feedback accuracy, and bang-bang controller deadband.

5.6 Determination of the Good Working Range of Speed and Angle

To determine the practical operating range of the servomotor, the experimental speed and position data were analyzed together. The objective was to identify the range of speed and angular displacement for which the servomotor exhibits low steady-state error and reliable positioning.

For the analysis, a percentage error of less than or equal to 5% is considered a low-error or good-working condition.

The percentage speed error is calculated as

$$\% \text{Speed Error} = \frac{|N_{meas} - N_{ref}|}{N_{ref}} \times 100$$

while the position error is calculated as

$$\% \text{Position Error} = \frac{|\theta_{obs} - \theta_{ref}|}{\theta_{ref}} \times 100$$

Therefore, the selected low-error criterion is

$$\% \text{Error} \leq 5\%$$

Speed-Control Performance

The speed-control experiments were performed in both forward and reverse directions. The corresponding percentage-error plots are shown in Figures 8 and 9.

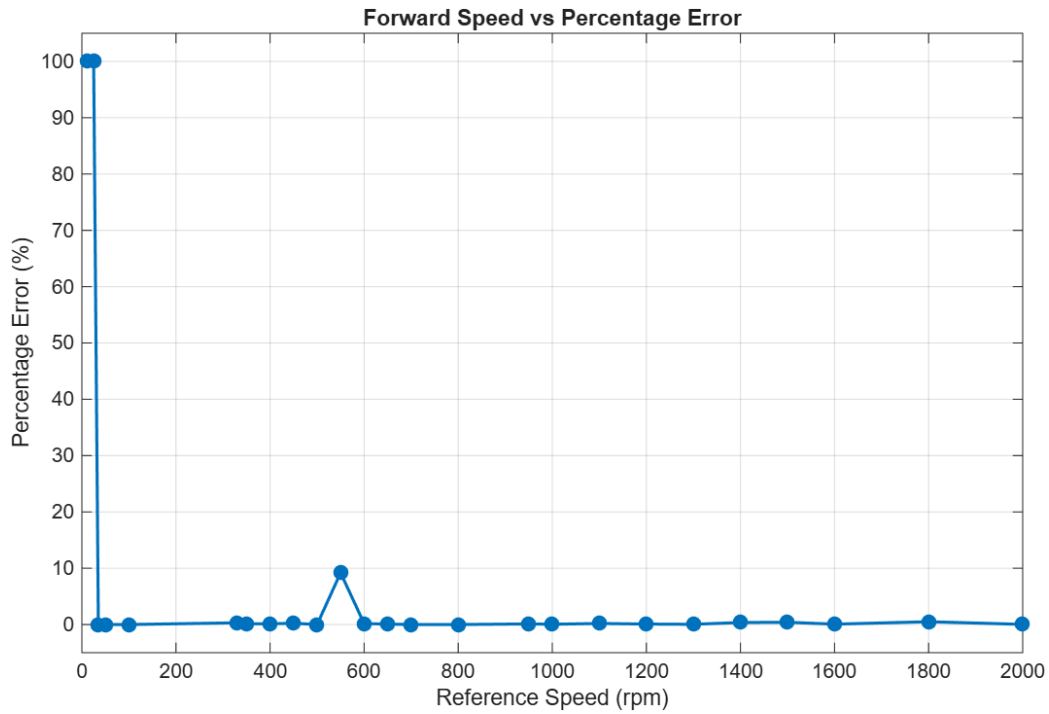


Figure 8: Forward speed versus percentage error.

The forward-speed graph shows that the speed error is very small over most of the normal operating range. At reference speeds from approximately 35 rpm up to 2000 rpm, the measured speed is generally close to the reference value. The major exceptions occur at extremely low speeds.

For example, at 11 rpm and 25 rpm, the measured speed was zero. Thus,

$$\%Error_{11} = |0 - 11| / 11 \times 100 = 100\%,$$

$$\%Error_{25} = |0 - 25| / 25 \times 100 = 100\%.$$

This indicates that the motor has a practical low-speed limitation. At very low reference speeds, the available motor torque and control action may not be sufficient to overcome static friction and maintain continuous rotation.

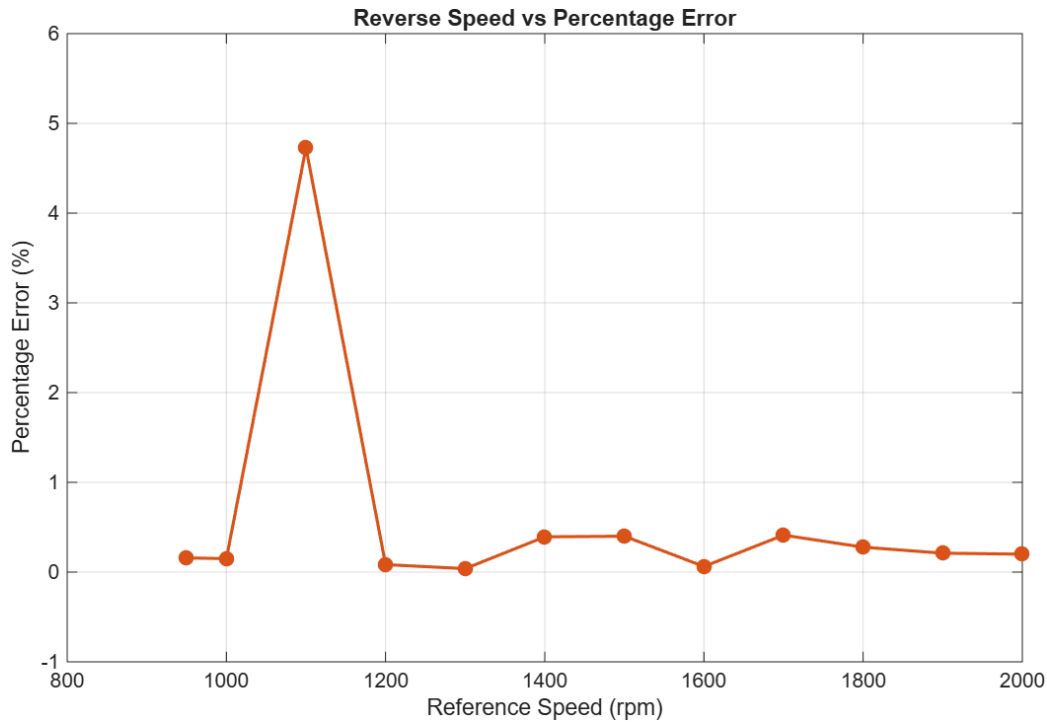


Figure 9: Reverse speed versus percentage error.

The reverse-speed graph also shows generally low error over the nominal operating range. Most of the measured points remain below approximately 1% error. An isolated larger deviation can be observed around 1100 rpm. This may be associated with experimental measurement variation, transient behaviour, or feedback fluctuations.

For example, at 2000 rpm, the measured reverse speed was approximately 1996 rpm:

$$eN = 1996 - 2000 = -4 \text{ rpm, giving } \%Error = 4/2000 \times 100 = 0.20\%.$$

Similarly, at 1800 rpm,

$$eN = 1805 - 1800 = +5 \text{ rpm, and therefore } \%Error = 5/1800 \times 100 \approx 0.28\%.$$

Thus, the speed-control results demonstrate that the servo motor provides good speed tracking in both directions over its normal operating range.

Position-Control Performance

The positioning performance was evaluated at constant speeds of 1800 rpm, 800 rpm, 100 rpm, and 50 rpm. Both forward and backward position errors were analyzed as a function of the commanded reference angle.

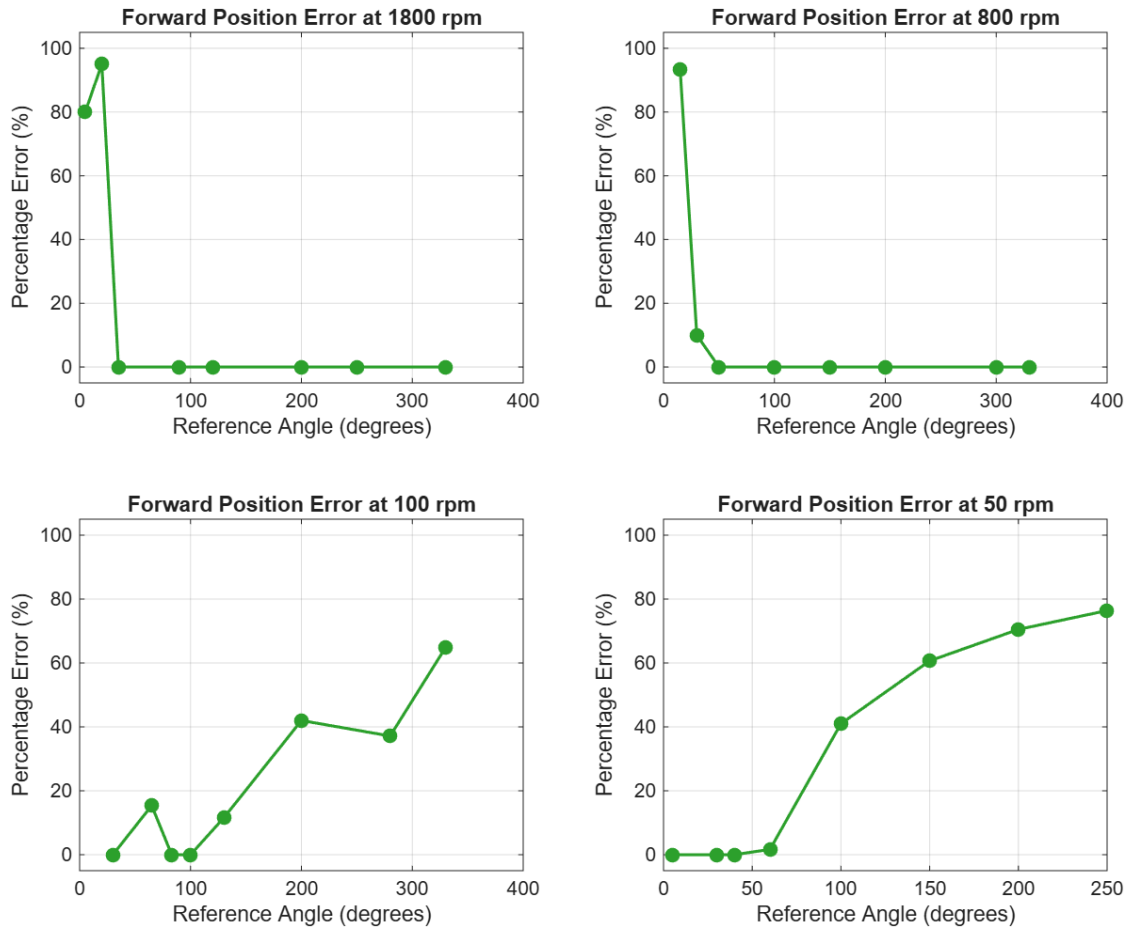


Figure 10: Forward position error as a function of reference angle at different operating speeds.

The forward-position results show a strong dependence of error on both speed and angular displacement.

At 1800 rpm, the small angular commands produce the largest errors. For example, $5^\circ \rightarrow 1^\circ$ gives $\%Error = |1-5|/5 \times 100 = 80\%$. Similarly, $20^\circ \rightarrow 1^\circ$ gives $\%Error = |1-20|/20 \times 100 = 95\%$.

However, larger angular commands such as 35° , 90° , 120° , 200° , 250° , and 330° show almost zero forward-position error at 1800 rpm.

This demonstrates that high speed does not necessarily produce high position error for every angular command. Instead, small-angle commands are particularly sensitive to the stopping dynamics of the motor.

At 800 rpm, the forward-position graph shows that most of the tested medium and large angular commands have negligible error. The angles 50° , 100° , 150° , 200° , 300° , 330° were achieved with approximately zero forward-position error.

However, the 15° command produced only 1° of observed motion, giving $\%Error = |1-15|/15 \times 100 = 93.33\%$.

Thus, even at 800 rpm, very small angular commands should be avoided when high positioning accuracy is required.

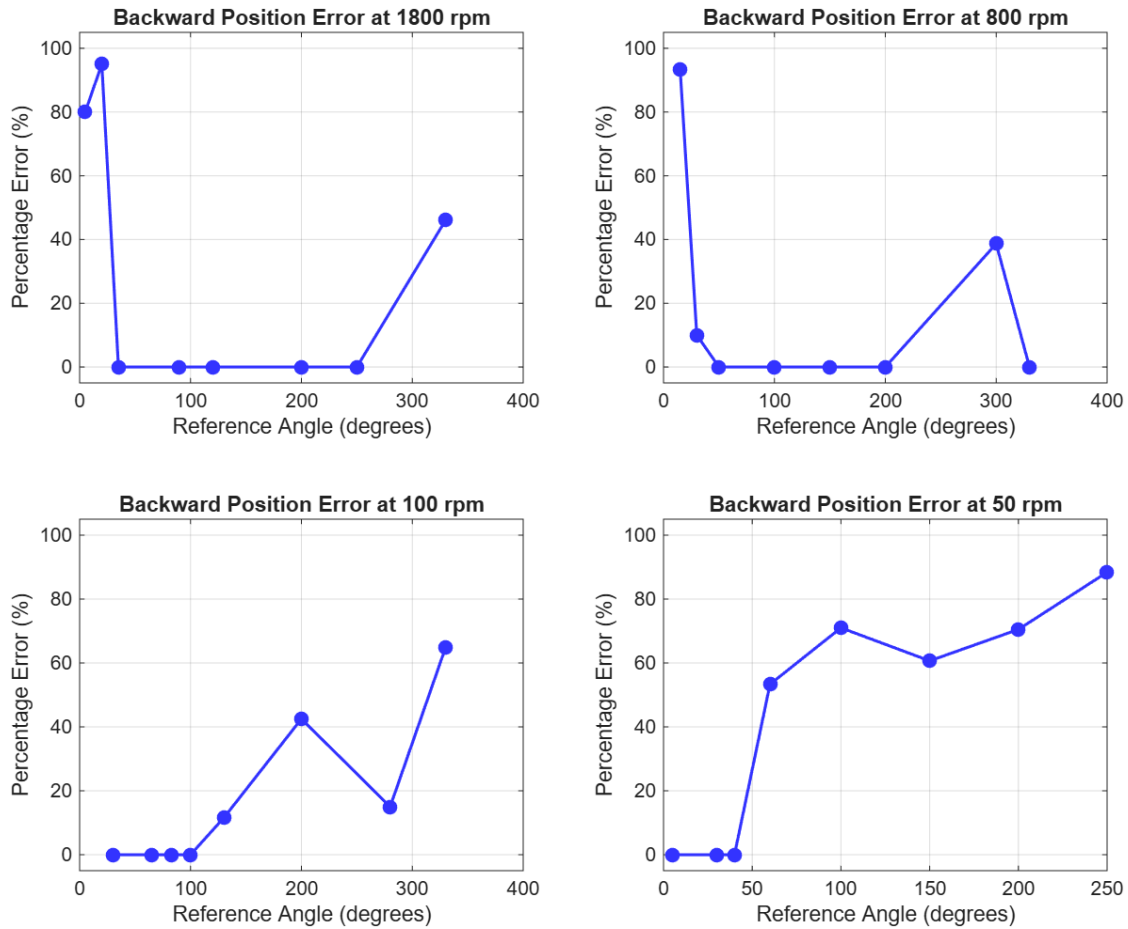


Figure 11: Backward position error as a function of reference angle at different operating speeds.

The backward-position results show a similar dependence on speed and angular displacement. At 1800 rpm, very small angles such as 5° and 20° produce errors of approximately 80% and 95%, respectively. At 800 rpm, the 15° command also produces approximately 93.3% error.

At lower speeds, particularly 50 rpm and 100 rpm, the error increases for several larger angular commands. This indicates that friction, stiction, and controller deadband become increasingly important at low operating speeds.

Effect of High Speed on Small Angular Commands

The high-speed experimental results show that small angular commands are particularly difficult to control accurately using bang-bang control.

The kinetic energy stored in the rotor is

$$E_k = \frac{1}{2}J\omega^2$$

where J is the rotor moment of inertia and ω is the angular speed.

Since $E_k \propto \omega^2$, the rotor possesses considerably more kinetic energy at high speed. When the bang-bang controller changes its output as the motor approaches the target, the rotor cannot stop instantaneously. The stored mechanical energy causes the shaft to continue moving during the stopping interval.

For small angular commands, the stopping distance can become comparable to the commanded displacement. Consequently, even a relatively small absolute positioning deviation produces a very large percentage error.

Therefore, the experimental results can be summarized as

High Speed + Small Angle → Large Positioning Error

particularly for the tested 5° , 15° , and 20° commands.

Effect of Low Speed and Friction

At low speeds, the rotor has considerably less kinetic energy. As a result, mechanical friction and static friction become relatively more significant.

When the position error enters the deadband of the bang-bang controller, the control action may be removed or changed. If the available motor torque is not sufficient to overcome static friction, the motor may stop before reaching the reference position.

For example, at 50 rpm, a 250° forward command resulted in an observed angle of only 59° :

$$e\theta = 59 - 250 = -191^\circ, \text{ and } \%Error = 191/250 \times 100 = 76.4\%.$$

Similarly, at 100 rpm, a 200° command resulted in an observed forward angle of 116° :

$$e\theta = 116 - 200 = -84^\circ, \text{ giving } \%Error = 84/200 \times 100 = 42\%.$$

These results indicate that low-speed operation can produce significant positioning errors because friction and deadband become dominant.

Effect of Angular Range

The experimental results also demonstrate that percentage error is strongly affected by the magnitude of the commanded angle.

For example, an absolute error of 2° corresponds to $2/20 \times 100 = 10\%$ for a 20° command, whereas the same 2° error corresponds to $2/90 \times 100 \approx 2.22\%$ for a 90° command.

Therefore, small angular commands naturally produce larger percentage errors for the same absolute positioning deviation.

The experimental results indicate that medium and large angular commands generally provide better percentage accuracy at moderate and high operating speeds, while very small angular commands are more sensitive to the stopping and feedback dynamics.

Backward Positioning and Low-Speed Drift

The backward-position plots show that the error can become particularly large at low speeds.

At 50 rpm, the backward-position error increases to approximately 50–90% for several larger angular commands. This indicates a strong influence of friction and insufficient corrective torque.

Similarly, at 100 rpm, the backward error reaches approximately 65% for the 330° command.

The system can also exhibit drift around the home or 0° position at low speeds. Since a pure bang-bang controller does not continuously apply a proportional corrective torque inside the deadband, small disturbances and friction can prevent the rotor from maintaining the exact reference position. Therefore,

Low Speed → Friction + Deadband → Position Drift

can explain the large errors observed in the low-speed backward-position measurements.

Determination of the Low-Error Working Region

Based on the experimental results, the condition $\%Error \leq 5\%$ was selected to identify a low-error operating point.

The tested operating points satisfying this criterion are summarized below.

Speed (rpm)	Reference Angle	Forward Position Performance
1800	35°, 90°, 120°, 200°, 250°, 330°	Approximately 0% error
800	50°, 100°, 150°, 200°, 300°, 330°	Approximately 0% error
100	30°, 83°, 100°	Approximately 0% error
50	5°, 30°, 40°	Approximately 0% error

Table 5: Low-error operating points obtained from the position-control experiment.

From the tested operating points, 800 rpm provides the most consistent low-error performance over a broad angular range. At this speed, all the tested angles from 50° to 330° listed in Table 5 produced approximately zero forward position error.

It is important to note that this does not prove that every angle between 50° and 330° will have less than 5% error because only discrete angular positions were tested. Therefore, the result should be reported as a tested operating region rather than a guaranteed continuous range.

Recommended Speed and Angle Range

Combining the speed and position results gives the following practical interpretation:

Speed	Suitable Tested Angles	Experimental Observation
1800 rpm	35°, 90°, 120°, 200°, 250°, 330°	Very good forward positioning for medium and large angles, but poor accuracy for small angles such as 5° and 20°.
800 rpm	50°, 100°, 150°, 200°, 300°, 330°	Best overall tested operating condition with approximately zero forward-position error at all listed angles.
100 rpm	30°, 83°, 100°	Good positioning only at selected angular commands; larger angles show significant errors.
50 rpm	5°, 30°, 40°	Limited operating region because friction and deadband cause large errors at larger angles.

Table 6: Recommended operating regions based on experimental results.

For speed control, the data indicate that the motor operates with generally low error over the nominal tested region of approximately $35 \text{ rpm} \leq N \leq 2000 \text{ rpm}$, although isolated measurements show larger deviations. The extreme low-speed region below approximately 35 rpm is not recommended because the motor failed to achieve the commanded speed at 11 rpm and 25 rpm.

For position control, the most consistent tested condition was $N \approx 800 \text{ rpm}$ with the tested angular commands 50°, 100°, 150°, 200°, 300°, 330°.

Hence, for the present experimental setup, approximately 800 rpm with medium-to-large angular commands provides the most reliable combination of speed and position accuracy.

Overall Conclusion from the Working-Range Analysis

The experimental results demonstrate that the optimum operating condition cannot be determined from speed alone. The positioning error is governed by the combined effects of motor speed, commanded angular displacement, mechanical inertia, friction, stiction, sensor feedback, control delay, and the deadband of the bang-bang controller.

The main observations are:

- The steady-state speed error is generally below 1% over most of the nominal operating range from approximately 35 rpm to 2000 rpm.
- At extremely low speeds such as 11 rpm and 25 rpm, the measured speed was zero, resulting in a 100% speed error.
- Small angular commands at high speed can produce very large positioning errors. At 1800 rpm, 5° and 20° commands produced approximately 80% and 95% error, respectively.
- At low speeds, friction, stiction, and controller deadband become dominant and can produce large errors even for relatively large angular commands.
- At 50 rpm, the 250° forward command produced approximately 76.4% error.
- At 800 rpm, the tested medium and large angular positions showed the most consistent low-error forward positioning performance.
- Therefore, based on the experimentally tested points, the preferred operating condition is approximately 800 rpm with medium-to-large angular commands.

Thus, the experimentally determined practical working region can be summarized as

Good Speed Tracking: 35–2000 rpm
Preferred Positioning Speed: $\approx$ 800 rpm
Preferred Tested Angles: 50°–330° (selected tested points)

This range represents the best performance observed in the present experiment and should not be considered a universal operating limit for the servomotor.

6. Applications of Control in Servomotors

Servomotors are widely used in applications where accurate control of position, speed, and angular displacement is required. Their closed-loop feedback mechanism allows the motor to continuously compare the desired position with the actual position and make corrective adjustments. Some important applications are described below.

6.1 Automated Flow Control Valves

Servomotors are used to precisely control automated flow-control valves. The motor rotates the valve to a specific angular position and maintains that position using feedback control. This allows accurate regulation of fluid or gas flow in industrial processes.

6.2 CNC Machinery and 3D Printers

Servomotors are widely used in CNC machines, milling machines, laser cutters, and 3D printers to control the motion of different axes. They drive mechanisms such as ball screws along the X, Y, and Z axes, providing accurate and repeatable positioning of the tool or print head.

6.3 Robotic Joints and Manipulators

Industrial robotic arms use servomotors to control their individual joints. Each motor provides controlled angular movement and maintains the required joint position through feedback. This enables robots to perform precise operations such as welding, pick-and-place, assembly, and material handling.

6.4 Aerospace and Defense Systems

Servomotors are used in aerospace and defense systems where high reliability and precise angular positioning are required. Applications include fly-by-wire flight-control surfaces, radar antenna steering, missile guidance mechanisms, and thrust-vectoring systems. The closed-loop nature of servo control allows these systems to respond accurately to commanded position changes.

6.5 Gimbaled Antenna Pointing

Servomotors are also used in satellite communication systems for precise antenna pointing. High-gain communication antennas must maintain a strict line-of-sight with terrestrial ground stations while the satellite moves along its orbit.

Two-axis gimbal mechanisms driven by servomotors allow the antenna to rotate in azimuth and elevation. Position feedback continuously measures the antenna orientation, while the controller generates corrective commands to maintain accurate alignment with the ground station.

Thus, servo control is essential for maintaining reliable communication links, particularly when very small angular positioning errors can cause a significant reduction in received signal strength.

7. Conclusion

The experiment successfully demonstrated the speed and position control of a Mitsubishi AC servomotor using a closed-loop feedback system and a bang-bang control strategy. The experimental results showed that the servomotor provides good speed tracking over the normal operating range, with speed errors generally remaining below 1%, while very low speeds showed significant errors due to friction and insufficient control action.

For position control, the results showed that **small angular commands at high speeds are highly sensitive to motor inertia, stopping delay, and feedback limitations**. At low speeds, **friction, stiction, and controller deadband became the dominant sources of positioning error**. Among the tested operating conditions, **approximately 800 rpm with medium-to-large angular commands provided the most consistent positioning accuracy**.

Overall, the experiment demonstrated that reliable servo positioning depends not only on speed regulation but also on the appropriate selection of the operating speed, angular displacement, feedback system, and controller deadband. The experimental analysis therefore highlights the **practical limitations of bang-bang control and the importance of selecting suitable operating conditions for accurate servo-motor control**.